THE CATHOLIC DIOCESE OF KIGOMA

ST. FRANCIS XAVIER SECONDARY SCHOOL



MIDTERM EXAMINATION AUGUST 2025

SUBJECT CODE: 033
DATE: Tuesday, 26th August 2025
TIME: 2:30Hours

SUBJECT: BIOLOGY

INSTRUCTIONS

1. This paper consists of three sections, namely **A**, **B** and **C**, with a total of **10 questions**.
2. Answer **all questions** in each section.
3. Section **A** has a total of **15 marks**, section **B** have a total of **70 marks** and section **C** has a total of **15 marks.**
4. All writing should be in **blue** or **black pen**, except for diagrams that must be drawn in pencil.
5. Communication devices and any unauthorized materials are not allowed in the examination room.
6. Write your **name** on every page of your answer booklet(s).

Question number	FOR EXAMINER'S USE ONLY	
	Scores	Examiner's Initials
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
TOTAL		

CONTACTS
0762-460-481

EMAIL
paulisaya058@gmail.com

ADDRESS
P.O.Box 43 Kasulu

SECTION A: 15Marks

1. For each of the items (i) - (x), choose the correct answer among the given alternatives and write its letter beside the item number in the answer booklet provided

(i) Which of the following organelles is involved in respiration?
- A. Cell membrane
- B. Nucleus
- C. Chloroplast
- D. Mitochondrion

(ii) The certain building in your school has the two doors which open outward, large windows, good water supply, light, and heat sources. Also has the tables where student stay when learning through experiments. The name of this building is_____
- A. Library
- B. Washrooms
- C. Laboratory
- D. Dining hall

(iii) The following group of organisms belong to the same kingdom;
- A. Bat, Whale, Amoeba.
- B. Euglena, Yeast, Liverwort.
- C. Grasshopper, Bat, Fish.
- D. Amoeba, Housefly, Mosquito.

(iv) Farmers in a certain village found their maize crops having yellow leaves with dead spots at the margin and tips. Which mineral do the maize crops lack?
- A. Calcium
- B. Sulphur
- C. Phosphorus
- D. Potassium

(v) Which statement is true about the virus?
- A. It is active outside the host cell and dormant inside it.
- B. It is active inside the host cell and dormant outside it.
- C. It carries all life processes when outside the host cell
- D. It has both RNA and DNA as genetic materials. .

(vi) The members of the same species can breed to produce viable offspring. Members from different species will not interbreed. The following breeds cannot interbreed to produce even sterile or infertile offspring **except:**
- A. Wolves and lion
- B. Horse and donkey
- C. Horse and cattle
- D. Duck and chicken

(vii) The following are methods involved when scientists study things systematically except
- A. Identifying the problem
- B. Guess the conclusion
- C. Formulating hypothesis
- D. Experimentation.

(viii) Which of the following is **not** a feature that helps pathogenic bacteria to spread diseases and infection?
- A. Fimbriae or pill
- B. Flagellate
- C. Their shapes
- D. Transformation

(ix) Which of the following labels indicates a substance which catch fire easily?
A. Corrosive
B. Flammable
C. Harmful
D. Oxidant

(x) Which of the following is a group of animal tissues?
A. Muscle, blood and bones
B. Brain, bone and liver
C. Skin, heart and brain
D. Muscle, liver and skin

QUESTION	i.	ii.	iii.	iv.	v.	vi.	vii.	viii.	ix.	x.
ANSWER										

2. Match the responses in List B with phrases in list A by writing the letter of the correct response from List B below the item number of List A in the table provided.

LIST A	LIST B
i. A small pore which allow gaseous exchange in and out of a leaf. ii. A water proof material which protect the leaf from injury and excessive water loss. iii. A part of a leaf with cells loosely parked to allow maximum gaseous exchange. iv. A part of a leaf which have abundant chlorophyll to maximize light absorption and photosynthesis. v. A broad part of a leaf which increase large surface area for absorption of light and carbondioxide.	A. Palisade mesophyll B. Spongy mesophyll C. Midrib D. Lamina E. Margin F. Cuticle G. Epidermis H. Stomata I. Apex J. Midrib

LIST A	i.	ii.	iii.	iv.	v.
LIST B					

SECTION B: 70Marks

Answer all questions from this section

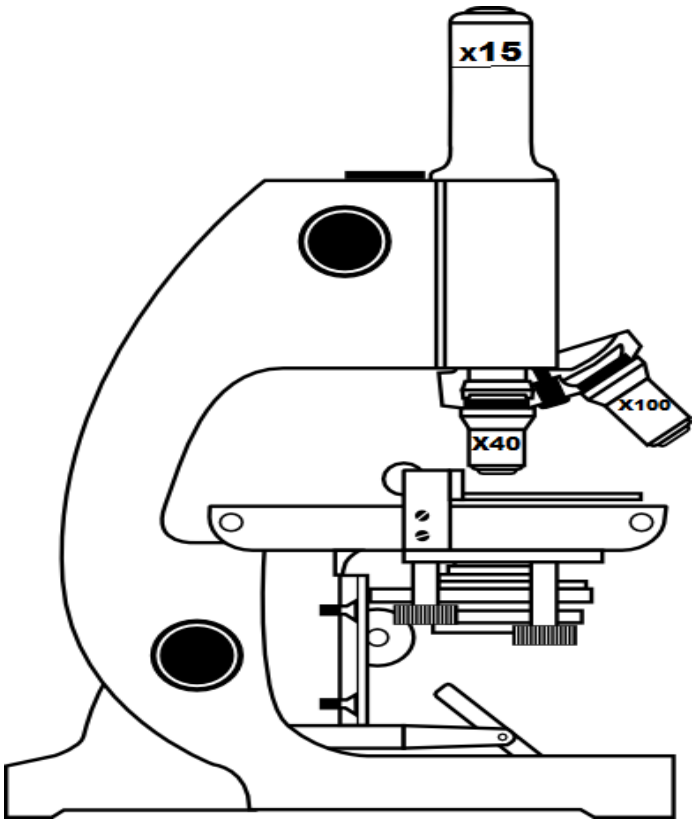
3. (a) Biology is more than a study of animals and plants. Explain by giving out three points.
(b) Outline four reasons as to why it is important to study Biology.
(c) When form one students were conducting private studies at night unknown organism entered the classroom. Such organism had **two pairs of legs**, **mouth**, **nose** and **eyes**. By using the only features observed on organism above identify and define three characteristics of living things that can be displayed by the organism that entered the classroom.

4. (a) Mr. David Edison was investigating the behavior of the form two students to complain about stomachache each day during the class hours. He used the scientific investigation skills. Briefly identify the **seven(7) steps** used by scientist during their investigation.
(b) Biology students assigned to observe the external features of a variety of preserved specimens.
i. Mention the sense organs that the students would primarily use and explain its function.

ii. Give one disadvantage of using sense organs to make observation

5. (a)Define the following terms as applied in Biology
   (i) Classification
   (ii) Binomial nomenclature
   (b) Mr. Lambe, wanted to order some garden plants and insects from India. He was required to write the names of those organisms using scientific names so that they can be easily read by scientists in India. Help Mr. Lambe the five rules he should follow in writing scientific names.
   (c) Follow the rules you have written in 4(a) to write the scientific names of the following Organisms
   (i) Human being
   (ii) Mango plant
   (iii) Housefly

6. (a) State the uses of each of the following laboratory apparatuses
   (i) Mortar and pestle
   (ii) Petri dish
   (iii) Test tube rack
   (iv) Bunsen burner
   (v) Thermometer
   (b) John attempted to observe plant cells using a prepared slide under a light microscope, but he was unable to obtain a clear image due to the following challenges.
   i. The slide containing specimen did not settle on the stage for observation.
   ii. He was unable to achieve sharp focus.
   iii. The image was too bright, making it difficult to see the detail of specimen.
   Educate your friend on how to address the challenges faced in order to obtain a clear image of the plant cell.
   (c) Observe carefully the figure and then calculate the total magnification of the specimen under observation.



7. (a) In 1665, Robert Hooke, an English scientist, designed a microscope and used it to Examine cork from the bark of a tree. He noticed that, the cork was made up of many box-like units which were packed up together to form a honeycomb. He called these units cells. State the constituents/components of his cell theory that were used by Robert Hooke. (Give two points).

(b) During an experiment that involved two sample **A** and **B**, the scientist observed cells whose organelles such as nucleus are enclosed in membranes in sample **A** and cells whose nuclear material are not bound with a membrane in sample **B**

(i) Identify types of cells observed in a sample **A** and **B** respectively

Sample A.....

Sample B.....

(ii) Write three (3) points that differentiate the types of cells observed in **sample A** and **B**

(c) (i) What do you understand by the term **"cell differentiation"**?

(ii) Give two examples of specialised cells in plants and two examples of specialized cells in a human body and then state their functions.

8. (a) During laboratory activities Juma soaked a piece of bread in water for few minutes Then he placed in a petri dish. Finally after one week he observed greyish substance with black tips have grown on a piece of bread.

(i) Name the organisms which were observed growing on a piece of bread.

(ii) Name the kingdom and phylum in which the organism named in (i) are placed.

(iii) State two characteristics of the phylum in which the organisms named in (i) belong.

(b) Draw a well labelled diagram of a fungi with the following features

i. Presence of cup –like structures called pileus (capsule) which contains the gills.

ii. Presence of a stem –like structure called stalk or stipe.

iii. Presence of volva which is found below the stem and the ring (annulus) which surrounds the stipe.

9. (a) What do you understand by the term **"photosynthesis"**?

(b) Outline the events which take place in each of the following stages of photosynthesis

i. Light dependent reaction

ii. Light independent reaction or dark stage

(c) Form one students at St. Francis Xavier secondary school used the following materials in the laboratory to perform an experiment shown in procedure **(a)** to **(g)** below

Materials: A leaf, ethanol or spirit, iodine solution,, source of heat, test tube , beaker, dropper, forceps, water, white tile.

Procedures

(a) They heated some water to boiling point in a beaker

(b) Then by using forceps they dipped the leaf in the hot water for about 40 seconds.

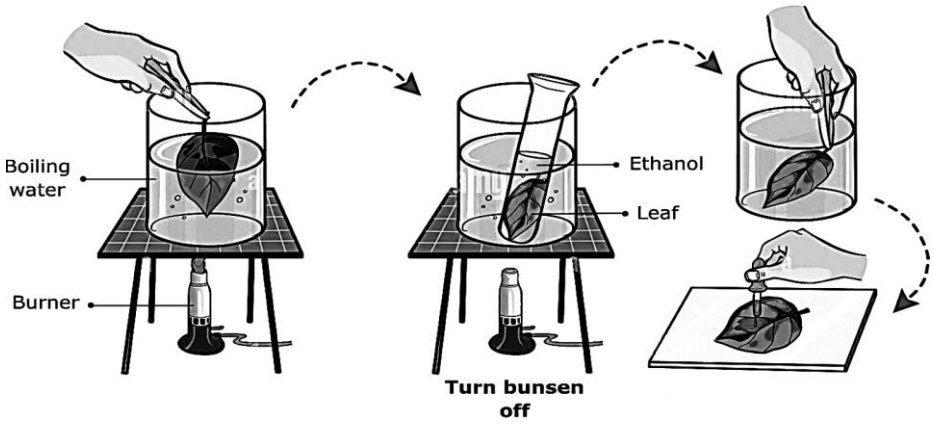
(c) They removed a leaf from the beaker.

(d) They placed a boiled leaf into the bottom of the test tube and covered it with alcohol(ethanol).

(e) Then they placed a tube in hot water until the alcohol boiled together with a leaf.

(f) Thereafter they removed a leaf from the test-tube containing ethanol and dipped it into hot water.

(g) Finally they spread the decolourized leaf on a white tile and drops of iodine solution on to it.



Questions

i. What was the aim of the experiment?
ii. Name the physiological process which was bemg investigated.
iii. Why was a leaf dipped in hot water for 40 seconds?
iv. Give reason, why a leaf was boiled in ethanol?
v. Why was a leaf dipped once again in hot water?
vi. Why was the alcohol boiled using a hot water bath?
vii. What observation did they make after applying iodine solution on a boiled and decolourised leaf?

SECTION C: 15Marks

10. Bacteria are one of the most important organisms in our daily life but few of them are also are harmful to our life. Support this statement by giving out four advantages and two disadvantages of bacteria in our daily life.